

TONGDAO, China

WMO: 578450

Lat: 26.1588N

Lon: 109.7777E

Elev: 397

StdP: 96.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.4	(c) -0.3	(d) -7.0	(e) 2.2	(f) 4.8	(g) -4.8	(h) 2.6	(i) 3.7	(j) 6.7	(k) 16.1	(l) 5.7	(m) 11.6	(n) 1.5	(o) 0	(p) 0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.2	(c) 33.7	(d) 24.4	(e) 32.7	(f) 24.2	(g) 31.6	(h) 24.0	(i) 26.0	(j) 30.6	(k) 25.5	(l) 29.8	(m) 25.1	(n) 29.2	(o) 2.9	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	21.0	27.6	24.5	20.4	27.2	24.1	19.9	27.0	83.4	30.6	81.3	30.0	79.5	29.3	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.5	5.5	4.7	DB	-3.2	35.9	1.2	0.9	-4.0	36.6	-4.7	37.1	-5.4	37.6	-6.2	38.2	
			WB	-3.7	26.8	1.1	1.5	-4.5	27.8	-5.2	28.7	-5.8	29.5	-6.6	30.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	6.1	8.5	12.5	18.1	22.0	25.2	27.1	26.8	23.8	18.8	13.6	8.2	
	DBStd	8.05	3.89	4.93	4.45	4.11	2.93	2.29	1.68	1.98	3.05	3.33	3.77	3.21	
	HDD10.0	322	132	82	24	1	0	0	0	0	0	0	10	73	
	HDD18.3	1401	378	277	188	54	4	0	0	0	2	34	147	314	
	CDD10.0	3099	12	39	101	244	373	454	531	521	414	274	119	17	
	CDD18.3	1136	0	2	7	47	119	204	272	262	166	50	6	1	
	CDH23.3	9041	0	8	36	280	775	1545	2559	2314	1219	273	30	1	
	CDH26.7	3067	0	1	3	50	209	441	935	917	452	57	1	0	
Wind	WSAvg	1.9	1.8	2.1	2.1	2.1	1.9	2.0	2.3	1.8	1.7	1.7	1.7	1.7	
Precipitation	PrecAvg	1417	56	67	96	171	236	231	143	135	75	90	76	43	
	PrecMax	1807	112	139	194	290	407	495	469	476	210	198	260	102	
	PrecMin	1006	8	13	41	67	75	61	29	20	3	12	4	2	
	PrecStd	221	25	31	36	60	93	99	103	101	52	49	54	27	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	24.7	26.7	29.8	32.1	33.3	34.8	35.0	33.8	30.6	26.1	21.2	
		MCWB	14.5	17.5	19.7	21.2	22.2	23.9	25.1	24.8	23.5	21.3	19.3	15.4	
	2%	DB	17.6	21.9	24.0	28.1	30.5	31.8	33.5	33.7	32.5	28.2	24.1	18.6	
		MCWB	12.9	16.2	18.2	20.7	21.7	23.7	24.8	24.6	23.4	20.4	17.5	12.4	
	5%	DB	14.8	19.6	21.9	26.5	29.0	30.5	32.4	32.7	31.2	26.5	22.2	16.0	
		MCWB	11.0	15.1	17.0	20.4	22.0	23.8	24.7	24.4	23.2	19.5	16.5	11.2	
	10%	DB	12.1	16.6	19.6	24.8	27.5	29.3	31.1	31.4	29.5	24.5	20.0	13.3	
		MCWB	9.3	13.1	16.0	20.0	21.9	23.9	24.5	24.3	22.6	18.9	15.6	9.5	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	18.2	20.6	23.2	25.1	26.0	26.6	26.7	25.5	23.6	20.9	17.6	
		MCDB	18.9	23.5	25.2	27.2	28.1	29.4	31.4	31.4	30.3	27.5	24.6	19.9	
	2%	WB	13.8	16.9	19.4	22.2	24.2	25.3	26.0	26.0	24.8	22.4	19.3	14.0	
		MCDB	16.7	20.9	23.2	26.2	27.7	28.9	30.7	31.0	29.4	25.9	22.5	17.1	
	5%	WB	11.4	15.6	18.0	21.5	23.5	24.9	25.5	25.5	24.2	21.2	17.7	11.7	
		MCDB	14.0	19.1	21.1	25.2	27.0	28.5	29.9	30.1	28.6	24.3	20.7	14.6	
	10%	WB	9.5	13.4	16.6	20.7	22.8	24.5	25.1	25.0	23.6	20.1	16.2	10.1	
		MCDB	11.6	16.2	18.8	24.0	26.0	27.8	29.2	29.2	27.7	23.2	18.6	12.6	
Mean Daily Temperature Range	MDBR	MDBR	5.5	6.0	6.5	7.2	7.5	6.5	7.2	8.1	8.7	7.8	7.9	7.2	
		MCDBR	12.5	11.3	11.8	11.1	11.7	9.4	10.1	10.5	11.6	11.9	12.5	13.9	
	5% DB	MCWBR	7.3	6.1	5.7	4.4	4.1	2.9	2.7	2.8	3.5	4.6	6.0	7.8	
		5% WB	MCDBR	9.4	9.6	8.5	7.9	7.4	6.5	8.2	8.8	9.1	8.4	8.7	10.6
		MCWBR	6.2	5.7	5.2	4.0	3.4	2.5	2.6	2.9	3.1	4.1	4.9	6.6	
		Clear-Sky Solar Irradiance	taub	0.622	0.643	0.737	0.706	0.599	0.576	0.540	0.601	0.633	0.686	0.603	0.608
			taud	1.604	1.611	1.483	1.588	1.857	1.939	2.041	1.853	1.746	1.581	1.704	1.649
			Ebn at Noon	613	647	617	657	731	744	770	722	682	606	626	601
Edh at Noon	232		247	296	272	208	190	172	206	224	251	208	214		
All-Sky Solar Radiation	RadAvg	1.56	1.92	2.42	3.08	3.51	3.59	4.75	4.48	3.81	2.98	2.41	1.91		
	RadStd	0.40	0.49	0.36	0.45	0.57	0.51	0.46	0.35	0.42	0.37	0.47	0.45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.26	N/A	N/A	+0.39	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.33	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	+118	+71

Nomenclature: See separate page